MultiLayerNetViz: A Multi-Layered Visualization for Information Diffusion in Social Networks

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In the past two decades, the study of the interplay between social structures and the information shared within them has generated a lot of interest. Notably in social networks, in which the variety of information spreading among agents is a predominant factor creating complex dynamics of opinion formation. Given the growing role of social media in shaping public discourse, this phenomenon calls for a deeper understanding of the structure of the underlying opinion landscape. In this work we introduce, `MultiLayerNetViz`[1] a lightweight customizable visualization framework to study and describe the socio-semantic structure of information diffusion in social networks. We show examples of this visualization on political data, given their salience in general questions of opinion formation, democracy and pluralism in the public sphere.

The proposed visualization is a Python module based on a coupled representation of the network of agents and of the topic/opinion space of the shared information. In fact, most social network studies with a semantic component lack a joint representation of the relationship between these two dimensions. Our approach, is based on an interaction model first introduced in [1], is relying on a multi-graph representation in which nodes (both agents and topics/opinions) are not only linked by their interactions, but are also spatially embedded in a coupled latent space. For instance, this design offers a clear view of specific semantic elements around which communities are formed, or ones that are at the core of the information. We illustrate our contribution with two case studies of political data: a physical social network the French Senate proposing amendments during 4 months of 2019[2] and a political opinion classification of tweets from the current or former Members of the French Parliament and web domains [3]. Despite the different nature and structure of theses datasets, both are based on the complex interplay between agents, around a variety of topics and opinions. Therefore, they are representative of the multifaceted nature of information diffusion in social networks.

In Fig. 1a is shown the temporal dynamics of proposed amendments, labeled by their date, senat group and topic. In Fig. 1b, the temporal data is then processed by the Mixture of Interacting Cascades (MIC) model [1] in order to capture the interaction between both topics and political groups. For the former, a structure with distinct branches is inferred: (Immigration, Liberties, Security) linked to (Europe, Foreign Policy) and opposed to (Education, Sport, Family). All these topics have a common center (Institution), which is the traditional core interest of the French Senate (as shown by the inter-layer edges). The Senate network is mostly organized around the biggest political group (LR), whose influence is the strongest on the other popular groups, whereas less mainstream ones (CRCE, NI) seem to have a more non-coordinated agenda. This provides evidence on the strategical synchronicity of these political actors regarding their democratic activities.

The data on the Twitter activity of (ex-)Members of the French Parliament (MPs) and web domains (Media Outlets) was already processed in [3] in which the political positions of these agents are calculated using the Chapel Hill Expert Survey [2]. This survey evaluates the position of political actors on a variety of fields that we refer to as "Ideologies". The visualization shown in Fig. 2 is based on the same approach as the one used for the Senate dataset, but here the shared information is characterized by an opinion. In this context, we combine and project the set of MPs and political web domains into a common latent political space built by a Principle Component Analysis (PCA) of their ideologies. Their projection into this new space is then used to spatialize their opinions. For clarity, we represent them into separate layers, on top of which we display the axes of ideologies projected into this latent political space. This representation is insightful for several reasons: it allows us to visualize the MPs and Media outlets in a common ideological space, highlighting the alignment between parties and media types (e.g. EELV and

[1]available on GitHub: https://github.com/gas-abel/MultiLayerNetViz

[2]provided by Magic Lemp (https://magic-lemp.com/fr/solutions/pluralism)

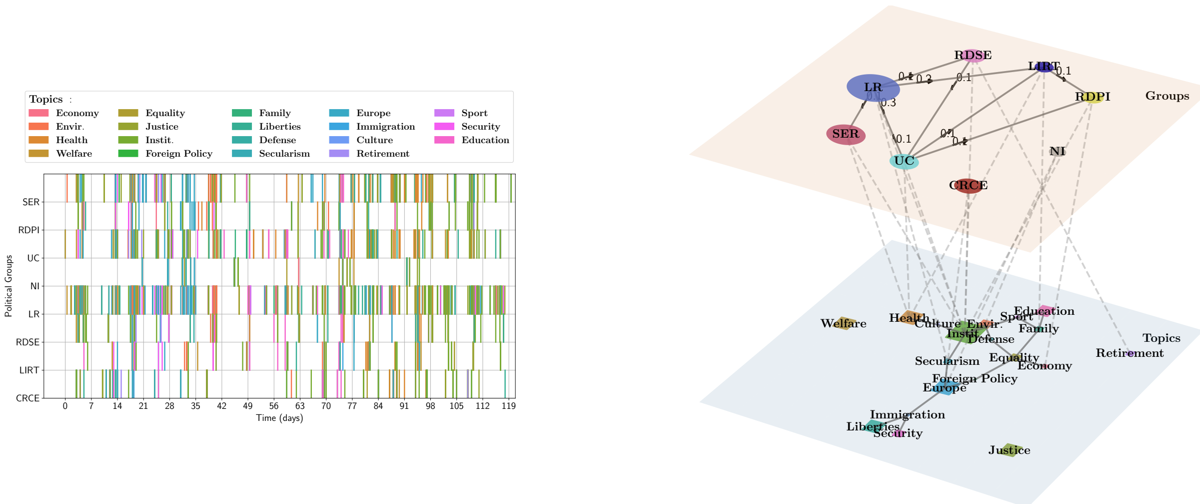

Figure 1: *Left*: Temporal unfolding of proposed amendments by 8 senate groups classified in 19 topics. *Right*: Layered Network Graph of this dataset represented by a fitted MIC model. Lower layer: set of corresponding topics represented as a graph built from the inferred interactions. Upper layer: social network projected onto the bill topics layout. Directional values displayed on the edges represent the influence between parties estimated by MIC. Node size represents the number of events.

PCF with Left Wing web domains), around specific ideologies (anti-elitism). It additionnaly sheds light on the most determinant political positions, whether they are orthogonal (EU position and left/right economy) or colinear (EU position and immigration).

In conclusion, `MultiLayerNetViz` offers an adaptable and insightful visualization framework for the representation of coupled social and semantic networks. It allows for a better understanding of the interplay between agents and topics or opinions, and can be applied to various situations that involve opinion formation and information diffusion.

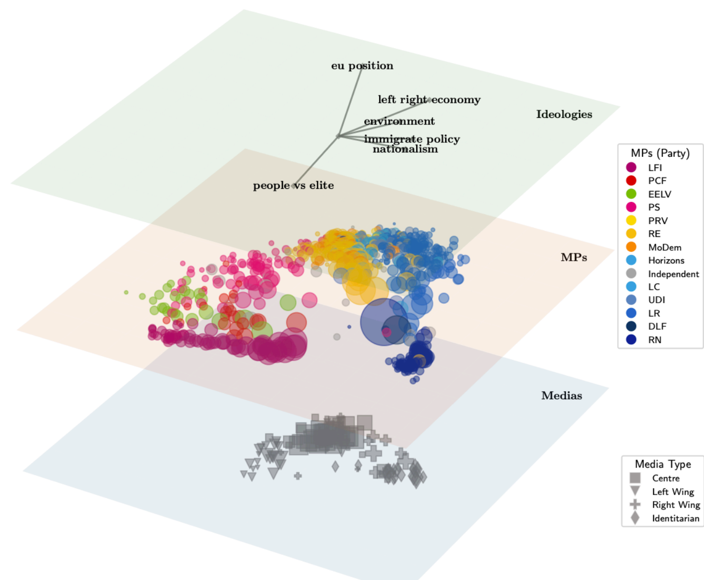

Figure 2: Visual representation of the 828 MPs and 400 Media outlets in the latent political space. Lower layer: Scatter plot of the political media outlets, markers represent media category (left legend), node size: number of events. Middle layer: Scatter plot of the MPs, colors represent the political party (right legend). Upper layer: Set of ideologies projected into the latent political space, vector norm reflects the relevance of these stances with respect to the principal components.

References

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